Chapter 14 (AIAMA)

Probabilistic Reasoning

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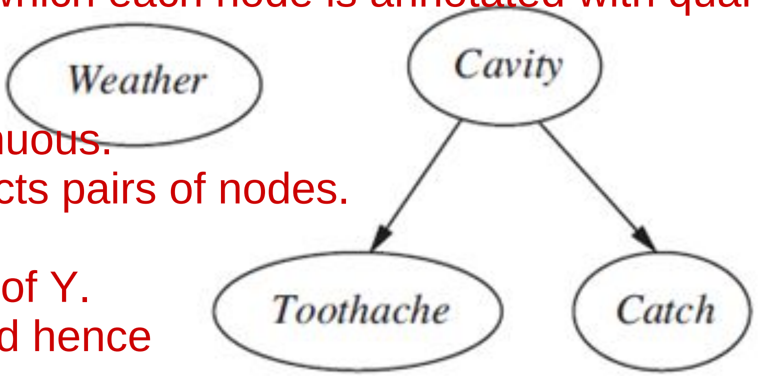
Bayesian Network

- A Bayesian network (BN) is a directed graph where each node is a random variable and edges show relationship (cause and effect) between the random variables.

A Bayesian network is a directed graph in which each node is annotated with quantitative probability information. The full specification is as follows:

1. Each node corresponds to a random variable, which may be discrete or continuous.
2. A set of directed links or arrows connects pairs of nodes. If there is an arrow from node X to node Y, X is said to be a parent of Y. The graph has no directed cycles (and hence is a directed acyclic graph, or DAG.
3. Each node Xi has a conditional probability distribution P(Xi |Parents(Xi)) that quantifies the effect of the parents on the node.



A simple Bayesian network in which *Weather* is independent of the other three variables and *Toothache* and *Catch* are conditionally independent, given *Cavity*

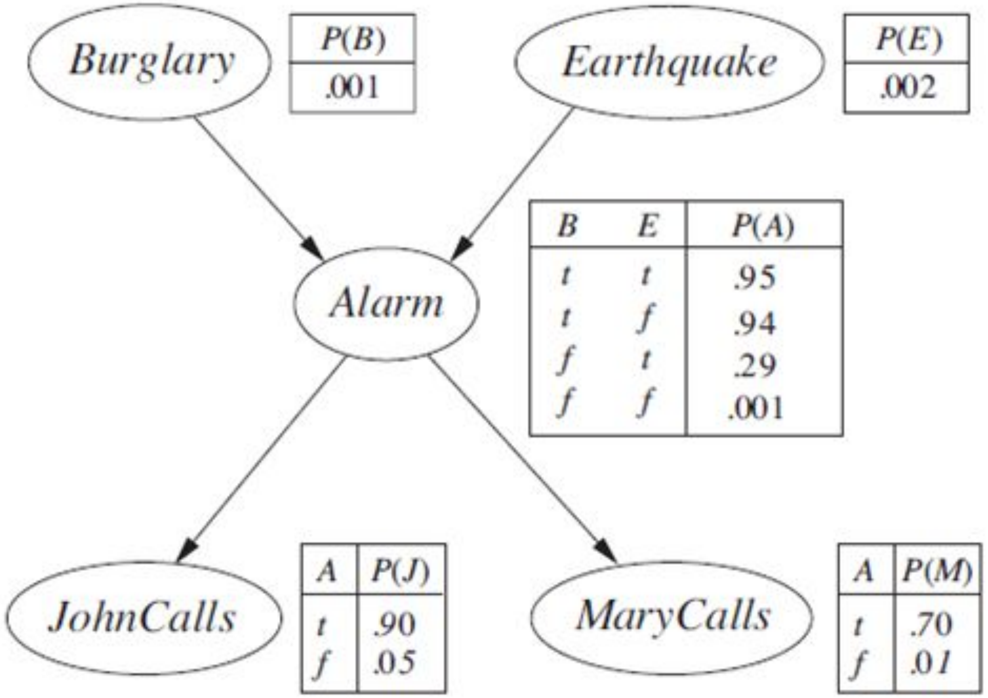
The topology of the network—the set of nodes and links—specifies the conditional independence relationships that hold in the domain, in a way that will be made precise shortly. The intuitive meaning of an arrow is typically that X has a direct influence on Y, which suggests that causes should be parents of effects.

Bayesian Network: Example

- You have a new burglar alarm installed at home. It is fairly reliable at detecting a burglary, but also responds on occasion to minor earthquakes.
- You also have two neighbors, John and Mary, who have promised to call you at work when they hear the alarm. John nearly always calls when he hears the alarm, but sometimes confuses the telephone ringing with the alarm and calls then, too.
- Mary, on the other hand, likes rather loud music and often misses the alarm altogether. Given the evidence of who has or has not called, we would like to estimate the probability of a burglary.

Bayesian Network: Example

- Bayesian network for burglary alarm Example

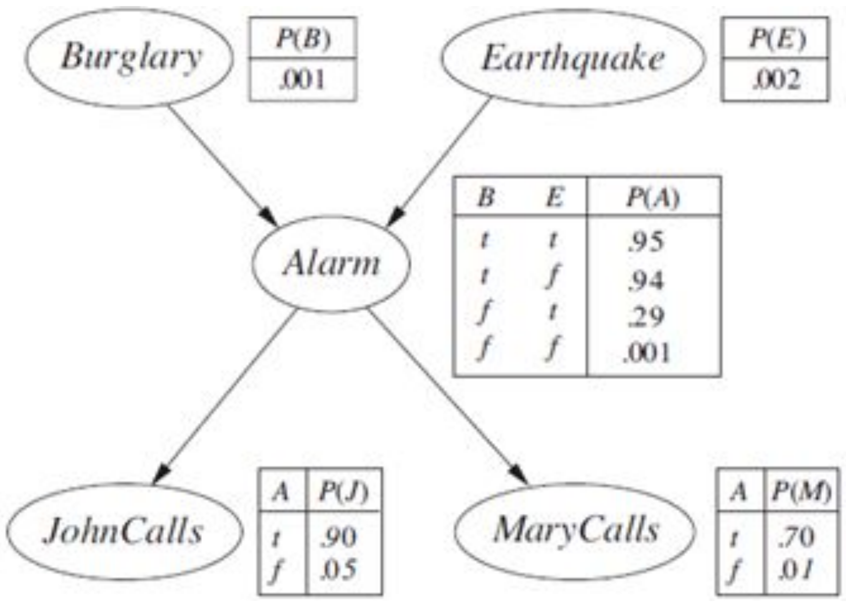


A typical Bayesian network, showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters B, E, A, J, and M stand for Burglary Earthquake, Alarm, JohnCalls, and MaryCalls, respectively.

Bayesian Network: Some Observations

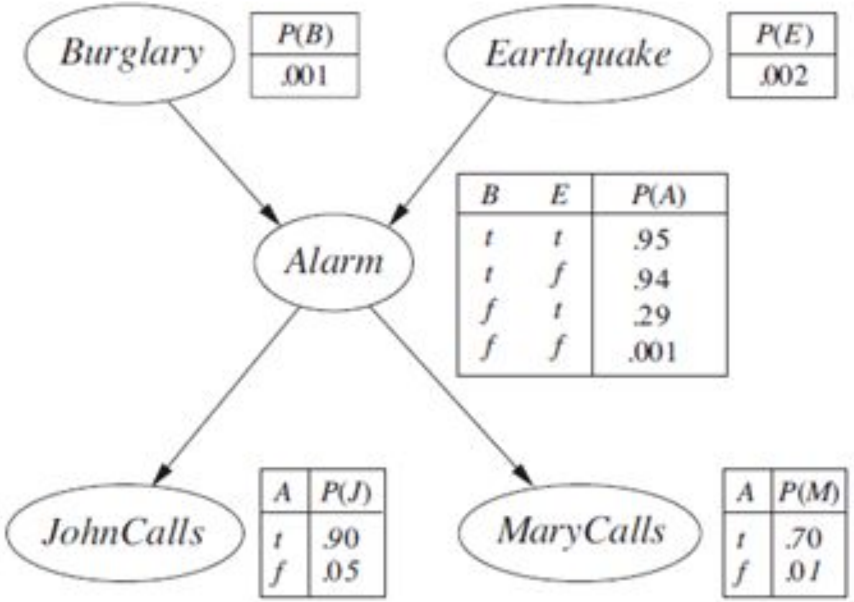
- Burglay and Earthquake directly affect the probability of Alarm going off
- John and Marry calls depending only on the Alarm (They know nothing about Burglary or Earthquake)

The network thus represents our assumptions that they do not perceive burglaries directly, they do not notice minor earth quakes, and they do not confer before calling.



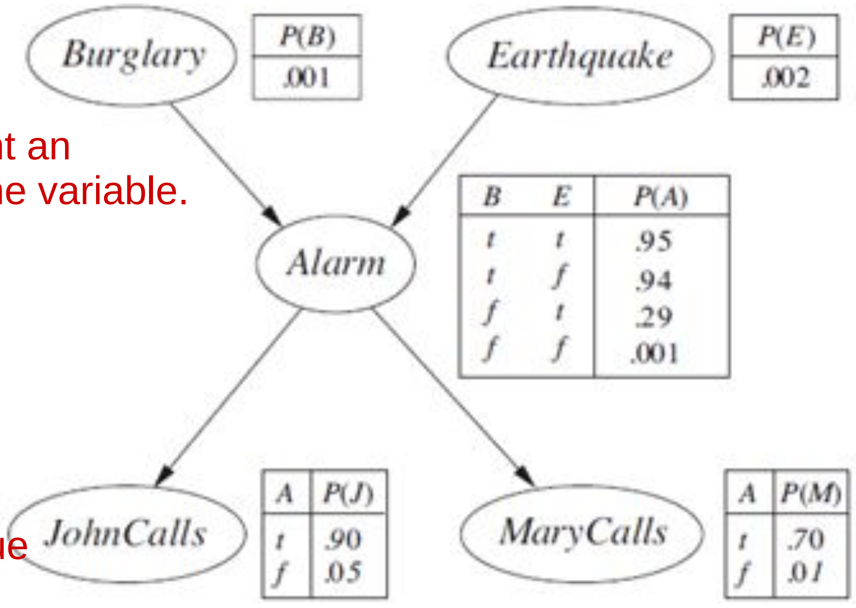
Bayesian Network: Example

- In a Bayesian network, each node is also annotated with **quantitative probability information** (conditional probability distribution).



Bayesian Network: Example

- Conditional probability distribution specified by a conditional probability table (CPT)
  - Each row specifies the probability of all values (of the node) given a combination of values of the parent nodes
  - Each row must sum to 1 because the entries represent an exhaustive set of cases for the variable.
  - For binary valued nodes, only one (1) probability value may be specified for each row. [as probabilities sum to 1]



For Boolean variables, once you know that the probability of a true value is p, the probability of false must be 1– p

In general, a table for a Boolean variable with k Boolean parents contains 2^k independently specifiable probabilities. A node with no parents has only one row, representing the prior probabilities of each possible value of the variable.

Bayesian Network: Size of CPTs

- A Boolean node have k Boolean parents. What will be the size of CPT for this node?

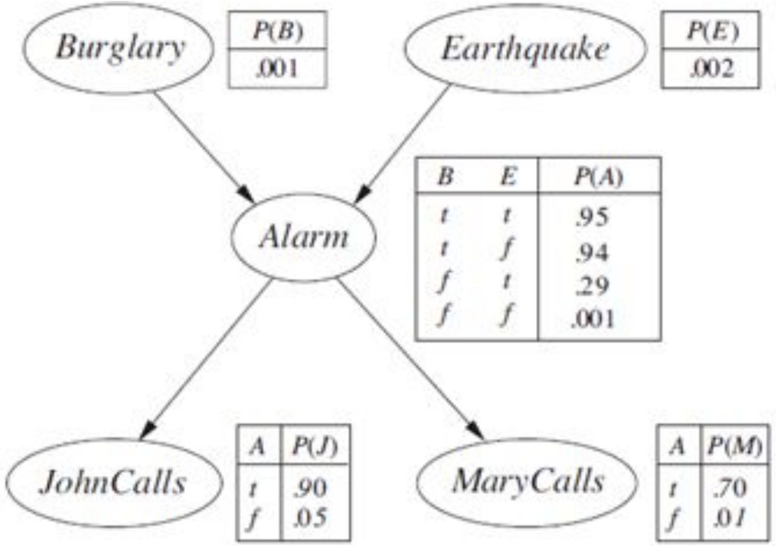
  Number of rows

  = possible combination of parent values

  = 2^k

  Number of entries

  = $(2^k)*1$ [Only one value need to be stored by row]

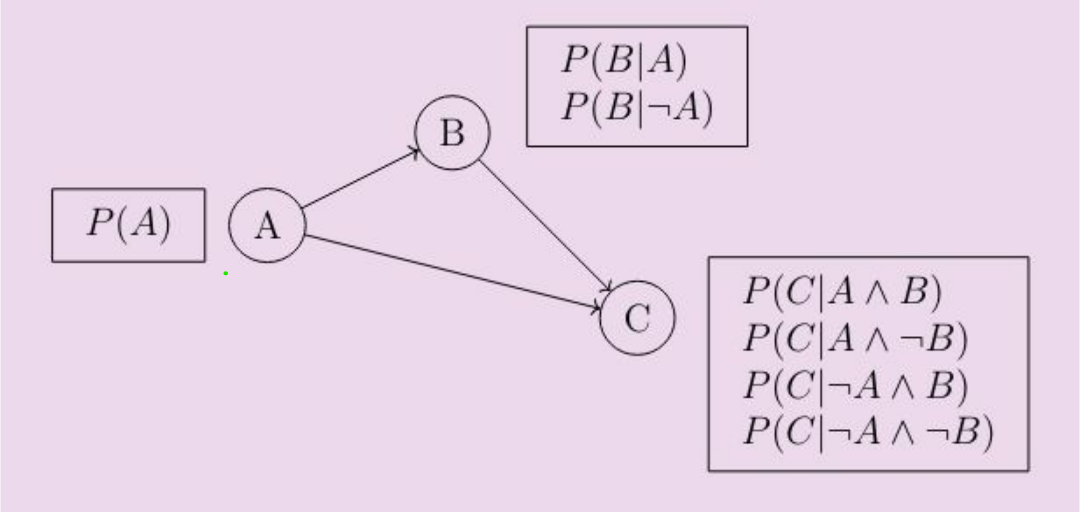


- How many probabilities are needed to represent the entire Bayesian Network?

  [Answer: 10] 2^2 + 2^1 + 2^1 + 2^0 + 2^0 = 10 (for Alarm, Burglary, Earthquake,JohnCalls, MaryCalls)

Bayesian Network: Size of CPTs

- How many probabilities are needed to represent the Bayesian Network? Assume all are Boolean variables. [Answer: 7] 2^0 + 2^1 + 2^2 = 7 (for A, B, C)



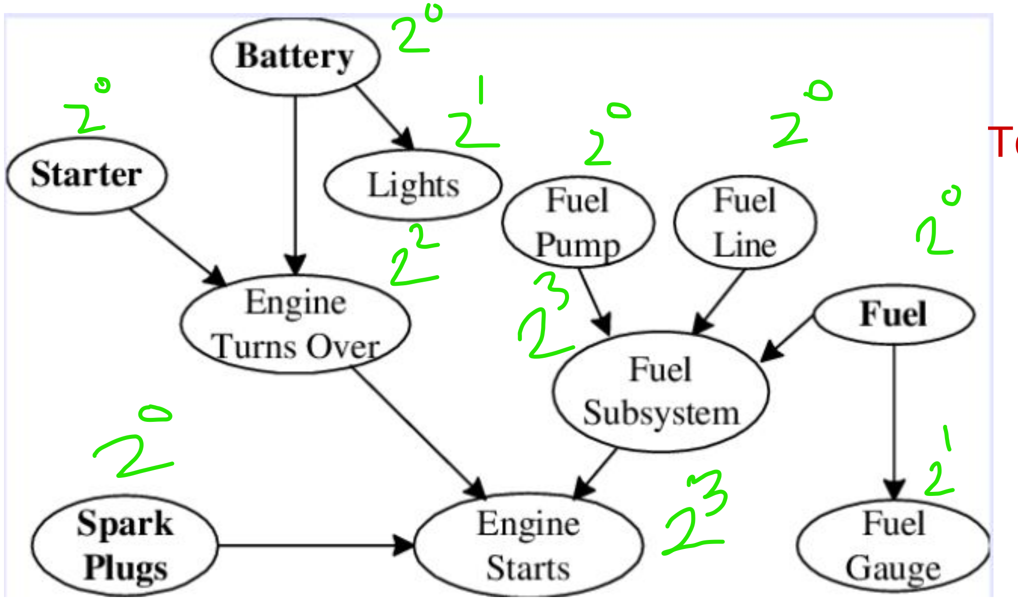
- How many will be needed if B and C are conditionally independent given A? [Answer : 5]

In this case, the conditional independence of B and C given A implies that P(C | A, B) = P(C | A)

2^0 + 2^1 + 2^1 = 5 (for A, B, C) Here, C is only dependent on A

Bayesian Network: More Example

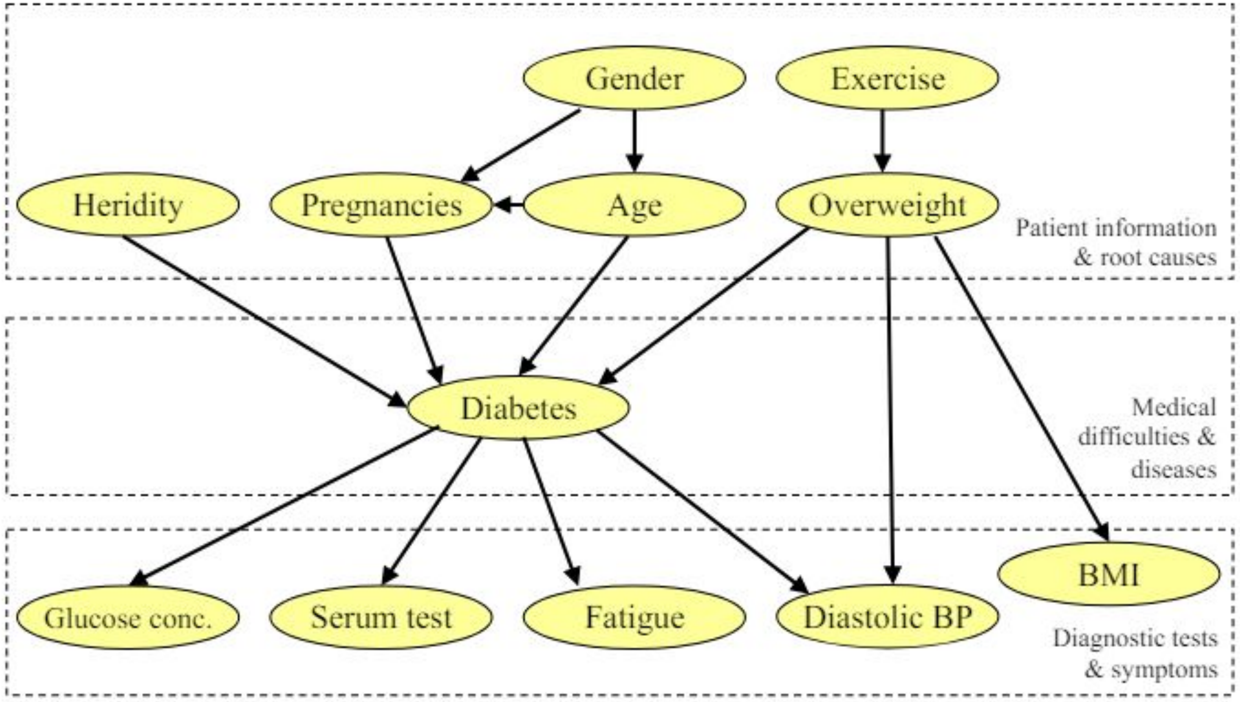
- A Bayesian Network for car diagnostic system.
  - we may have problems in a particular part of the car, and that may be caused by many different reasons.
  - The battery might cause the lights to have different problems or it may cause the engine to have problems, and that in turn may affect whether or not the engine can start.



Total probabilities needed = 30

Bayesian Network: More Example

- A Bayesian Network for diabetes diagnosis.
  - Symptoms and tests
  - Patient information: the patient's age, weight, medical history, genetics, and so on.



Bayesian Network: Full Specification

1. Each node corresponds to a random variable, which may be discrete or continuous.
2. Set of directed links or arrows connects pairs of nodes.
   - If there is an arrow from node X to node Y , X is said to be a parent of Y. The graph has no directed cycles and hence is a directed acyclic graph, or DAG.
3. Each node X_i has a conditional probability distribution $P(X_i \mid Parents(X_i))$ that quantifies the effect of the parents on the node.

Bayesian Network: Conditional Independence

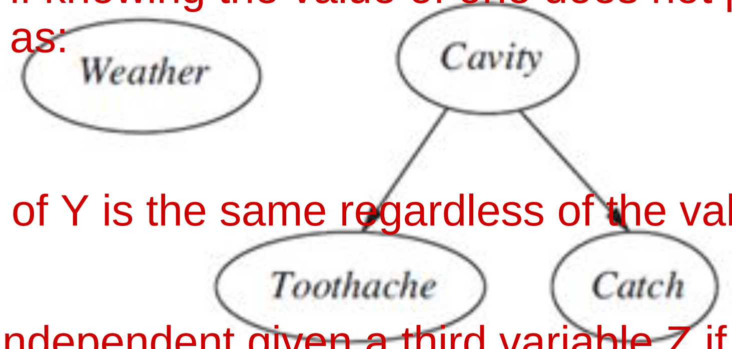
- Implicitly defines conditional independence of random variables

  - *Weather* is Independent of other variables

  - *Toothache* and *Catch* are conditionally independent given *Cavity*

Independence :
Two variables X and Y are independent if knowing the value of one does not provide any information about the value of the other. Mathematically, this is expressed as:

$P(X,Y) = P(X)\,P(Y)$
or, $P(Y|X) = P(Y)$



In other words, the probability distribution of Y is the same regardless of the value of X, and vice versa.

Conditional Independence :
Two variables X and Y are conditionally independent given a third variable Z if, once Z is known, X and Y become independent of each other. Mathematically, this is expressed as:

$P(X,Y|Z) = P(X|Z)\,P(Y|Z)$
or, $P(Y|X,Z) = P(Y|Z)$

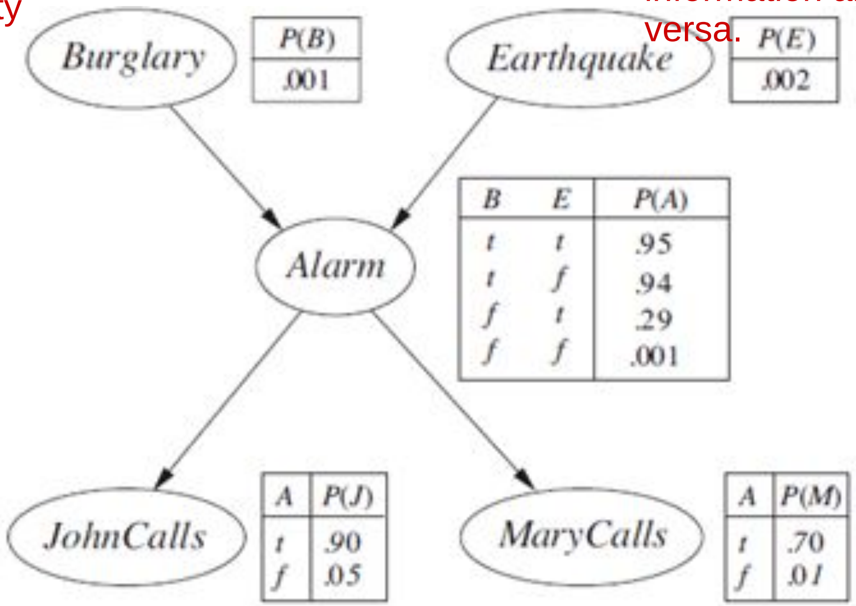
This means that if you know the value of Z, then knowing the value of X gives you no additional information about Y, and vice versa.

Bayesian Network: Conditional Independence

- Implicitly defines conditional independence of random variables
  - JohnCalls is Independent of MarryCalls given Alarm
  - MarryCalls is independent of EarthQuake given Alarm
  - Is Burglary independent of Earthquake? [Yes, Why?]

Independence in Bayesian Networks: Two variables are independent if there is no direct or indirect causal link between them without conditioning on other variables.
Structure of the Network: In the diagram, Burglary and Earthquake are both parent nodes of the Alarm node. This means they influence the Alarm, but there is no direct connection between Burglary and Earthquake.
No Causal Link: The lack of a direct or indirect causal link between Burglary and Earthquake means that knowing whether an earthquake has occurred doesn't give you any additional information about whether a burglary has occurred, and vice versa.

There are two ways in which one can understand the semantics of Bayesian networks. The first is to see the network as a representation of the joint probability distribution. The second is to view it as an encoding of a collection of conditional independence statements. The two views are equivalent, but the first turns out to be helpful in understanding how to construct networks, whereas the second is helpful in designing inference procedures.



Bayesian Network: Conditional Independence

- Implicitly defines conditional independence of random variables

  - Are Burglary and Watson independent? [No]

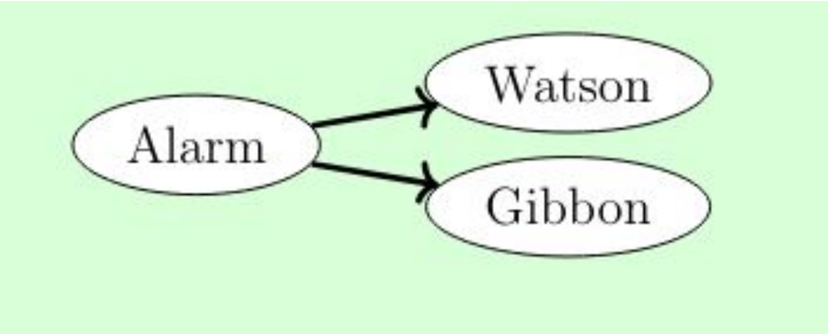
  - Are Burglary and Watson independent given Alarm? [Yes]

Bayesian Network: Conditional Independence

- Implicitly defines conditional independence of random variables

  - Are Watson and Gibbon independent? [No] Conditionally Independent

  - If Watson is calling, it is more likely that Alarm is off, which means that Gibbon is more likely to call



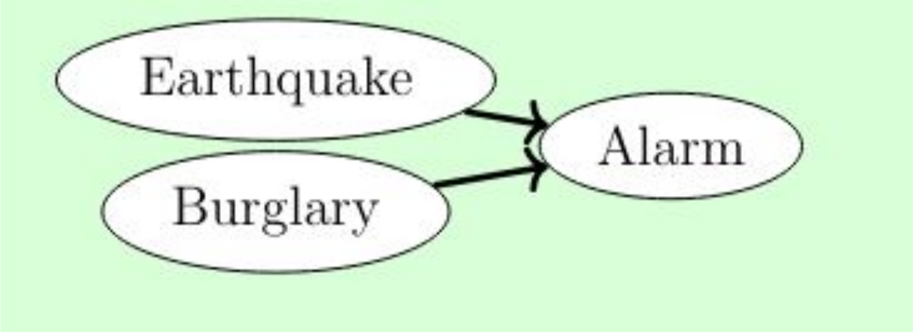
Bayesian Network: Conditional Independence

- Implicitly defines conditional independence of random variables

  - Are Earthquake and Burglary independent [Yes]

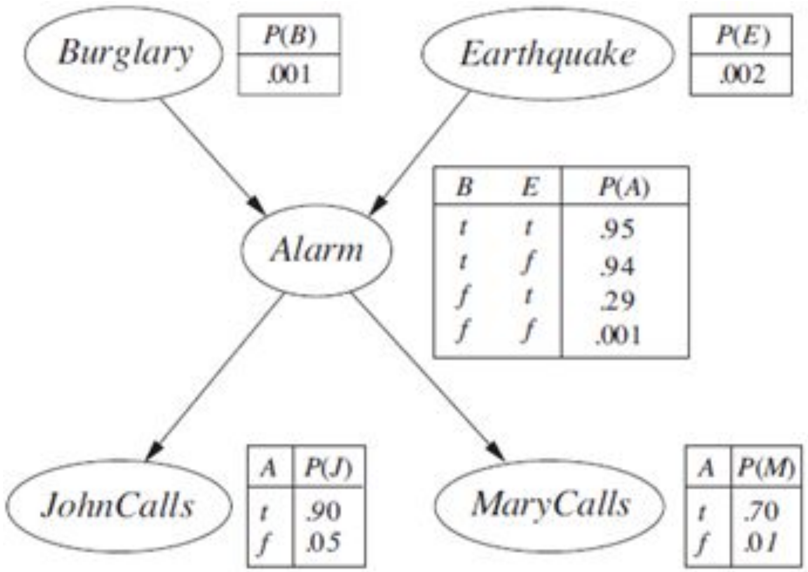
  - Are Earthquake and Burglary independent given Alarm [No]

  [If Earthquake is happening, then it is less likely that the Alarm is caused by Burglary.]



Bayesian Network: Full Joint Distribution

- *Bayesian network* + *conditional probability distribution* for each variable, given its parents specify full joint distribution for all the variables.



Bayesian Network: Full Joint Distribution

-
- Given a Bayesian Network with variables/nodes $x_1, x_2, \ldots, x_n$
- Find the joint probability $P(x_1, x_2, \ldots, x_n)$
- We can show that:

$$
\begin{aligned}
P(x_1, \ldots, x_n) &= P(x_n \,|\, x_{n-1}, \ldots, x_1) P(x_{n-1}, \ldots, x_1) \,. \\
&= P(x_n \,|\, x_{n-1}, \ldots, x_1) P(x_{n-1} \,|\, x_{n-2}, \ldots, x_1) \cdots P(x_2 \,|\, x_1) P(x_1) \\
&= \boxed{\prod_{i=1}^{n} P(x_i \,|\, x_{i-1}, \ldots, x_1) \,.}
\end{aligned}
$$

P(A | B) = P(A,B) / P(B)

Bayesian Network: Full Joint Distribution

- If nodes are ordered such that all parent nodes of $x_i \in \{x_{i-1} \ldots, x_1\}$, then the full joint distribution can be written as:

$$\boxed{P(x_1, x_2, \ldots, x_n) = \prod_{i=1}^{n} P(x_i | Parent(X_i))}$$

- We just need to find a topological sorting of the nodes for the above equation to become correct!

Bayesian Network: Compute Full Joint Distribution

- Calculate the probability that the alarm has sounded, but neither a burglary nor an earthquake has occurred, and both John and Mary call.

We have following values:

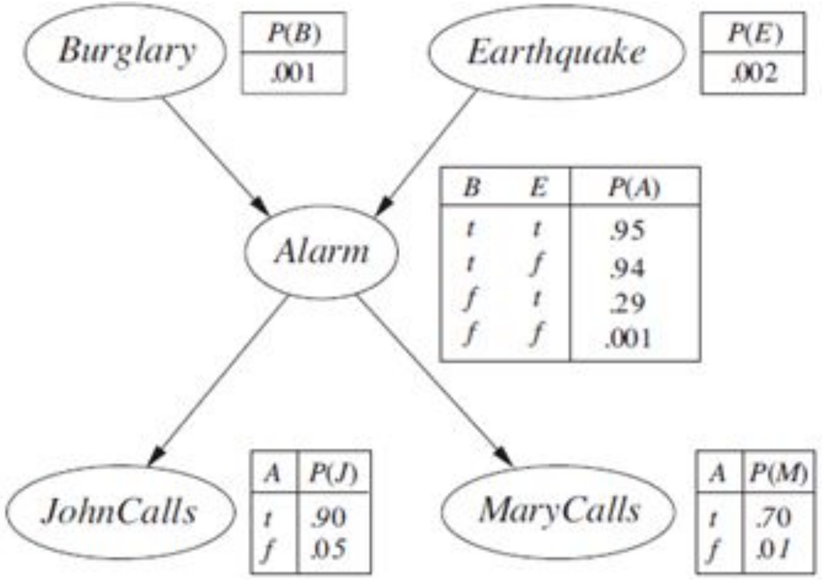
a = alarm sounds

$\neg b$ = no burglary

$\neg e$ = no earthquake

j = John calls

m = Marry calls

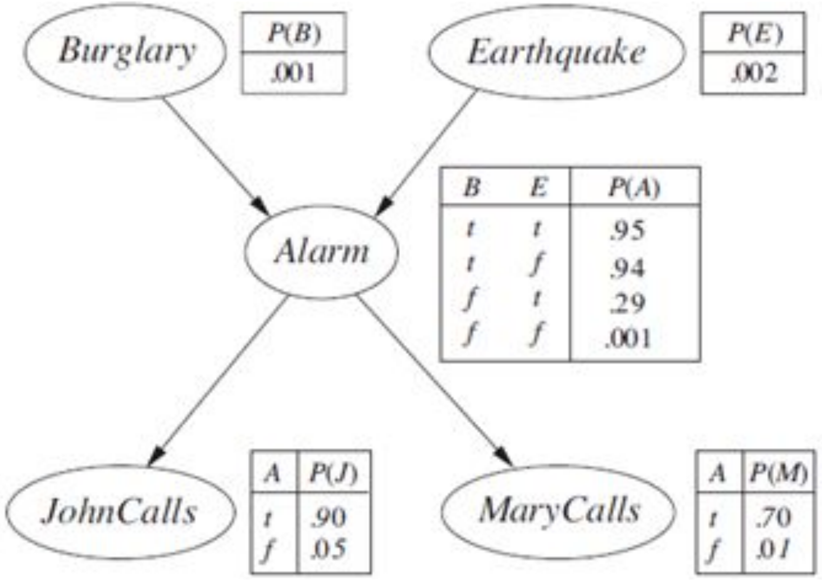


Bayesian Network: Compute Full Joint Distribution

- Calculate the probability that the alarm has sounded, but neither a burglary nor an earthquake has occurred, and both John and Mary call.

- We need to find: $P(a, \neg b, \neg e, j, m)$

- We can compute the full joint distribution using CPTs.



Bayesian Network: Compute Full Joint Distribution

P(MaryCalls| JohnCalls,Alarm,Earthquake,Burglary)= P(MaryCalls | Alarm) .
Thus, Alarm will be the only parent node for MaryCalls.

- $P(a, \neg b, \neg e, j, m)$

$= P(j, m, a, \neg b, \neg e)$ [rearrange as per topological sorting]

$= P(j|m, a, \neg b, \neg e)P(m, a, \neg b, \neg e)$ $[P(a, b) = P(a|b)P(b)]$

$= P(j|a)P(m, a, \neg b, \neg e)$ [j is independent of other variables given a]

$= P(j|a)P(m|a)P(a, \neg b, \neg e)$ [Same logic]

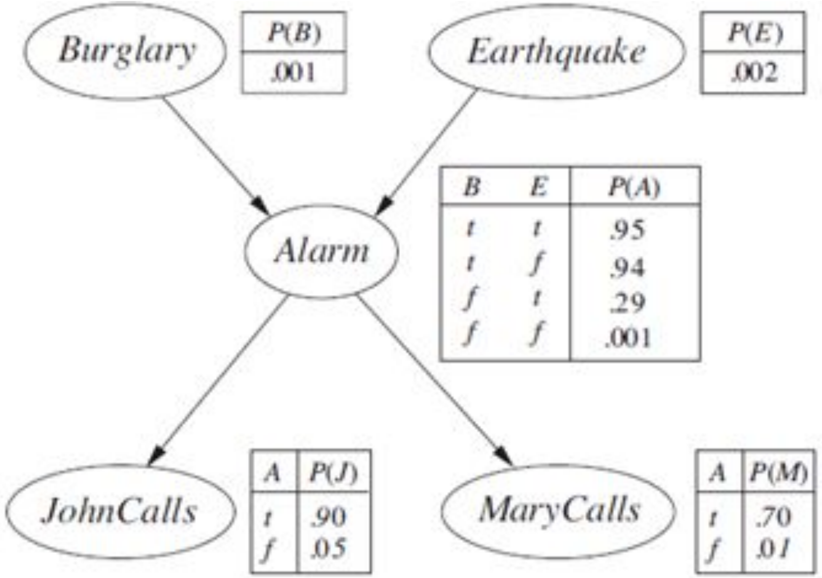
$= P(j|a)P(m|a)P(a|\neg b, \neg e)P(\neg b, \neg e)$

$= P(j|a)P(m|a)P(a|\neg b, \neg e)P(\neg b)P(\neg e)$

[b and e are independent of each other]

$= 0.9 * 0.7 * 0.001 * 0.999 * 0.998$

$= 0.000628$



P(B)
.001

P(E)
.002

B	E	P(A)
t	t	.95
t	f	.94
f	t	.29
f	f	.001

A	P(J)
t	.90
f	.05

A	P(M)
t	.70
f	.01

Chapter 14 (AIAMA)

Probabilistic Reasoning

Given a joint probability distribution over variables a set of variables X = X1, X2,..., Xn, we can make inferences of the form (Y | Z), where Y⊂X is the set of query variables, Z⊂X are the evidence variables. The other variables H (those not mentioned in the query) are called hidden variables. (Just For concept, ignore the Notations X,Y,Z,H)

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The basic task for any probabilistic inference system is to compute the posterior probability distribution for a set of query variables, given some observed event—that is, some assignment of values to a set of evidence variables. To simplify the presentation, we will consider only one query variable at a time; the algorithms can easily be extended to queries with multiple variables. We will use notations as follow : X denotes the query variable; E denotes the set of evidence variables E1,...,Em and e is a particular observed event; Y will denotes the nonevidence, non query variables Y1,...,Yl (called the hidden variables). Thus, the complete set of variables is X= {X}∪E∪Y. A typical query asks for the posterior probability distribution P(X |e).

Bayesian Network: Exact Inference

- **Consider the query**: $P(Burglary|JohnCalls = ture, MaryCalls = true)$
- We need to compute: $P(B|j,m)$ [compute for both b and $\neg b$]
- Express in terms of joint distributions [we already know how to compute joint distribution]

$$P(B|j,m) = \frac{P(B,j,m)}{P(j,m)} = \alpha P(B,j,m)$$

where α is the normalizing constant [we can find the value of α later]

Bayesian Network: Exact Inference

- How to compute $P(B, j, m)$?

- Add all hidden variables to get full joint distribution

$$P(B, j, m) = \sum_a \sum_e P(B, j, m, a, e) = \sum_a \sum_e P(j|a)P(m|a)P(a|B, e)P(B)P(e)$$

  - Query variable: B
  - Evidence variables: J, M
  - Hidden variables: A, E

Bayesian Network: Exact Inference

- Finally, we get the following:

  $P(B|j,m) = \alpha \sum_a \sum_e P(j|a)P(m|a)P(a|B,e)P(B)P(e)$

  - Compute $\mathrm{P}(b|j,m)$ and $\mathrm{P}(\neg b|j,m)$ [expression has α]
  - Find α [Using the eq. $\mathrm{P}(b|j,m) + \mathrm{P}(\neg b|j,m) = 1$]

- **Question**: How many operations (multiplications + additions)?
  - Answer: [For each $P(B|j,m)$] Mult: 16, Sum: 4, Total = 20 operations

Bayesian Network: Exact Inference

- Computing using join distribution naively:

$$P(B|j,m) = \alpha \sum_a \sum_e P(j|a)P(m|a)P(a|B,e)P(B)P(e)$$

- Required operations: 20
- **Can we improve?**
  - Yes, move sums inside [*closest to the factors having the hidden variable*]
  - Sums are evaluated early and gets multiplied once [*rather than every time during joint calculation*]

Bayesian Network: Exact Inference

- Move summations inside: $P(B|j,m) = \alpha \sum_a \sum_e P(j|a)P(m|a)P(a|B,e)P(B)P(e)$
- Two alternate ways to do it:
  - Option 1: $P(B|j,m) = \alpha P(B) \sum_a P(j|a)P(m|a) \sum_e P(a|B,e)P\ (e)$
  - Option 2: $P(B|j,m) = \alpha P(B) \sum_e P\ (e) \sum_a P(a|B,e)P(j|a)P(m|a)$
  - Which one is more efficient in terms of operations?

Bayesian Network: Exact Inference

- **Which one is more efficient?** Assume $B = true$ and compute both.
- Equation 1: $P(B|j,m) = \alpha P(B) \sum_a P(j|a) P(m|a) \sum_e P(a|B,e) P\ (e)$

  - Required # of multiplications: 9, # of sums = 3, total = 12

- Equation 2: $P(B|j,m) = \alpha P(B) \sum_e P\ (e) \sum_a P(a|B,e) P(j|a) P(m|a)$

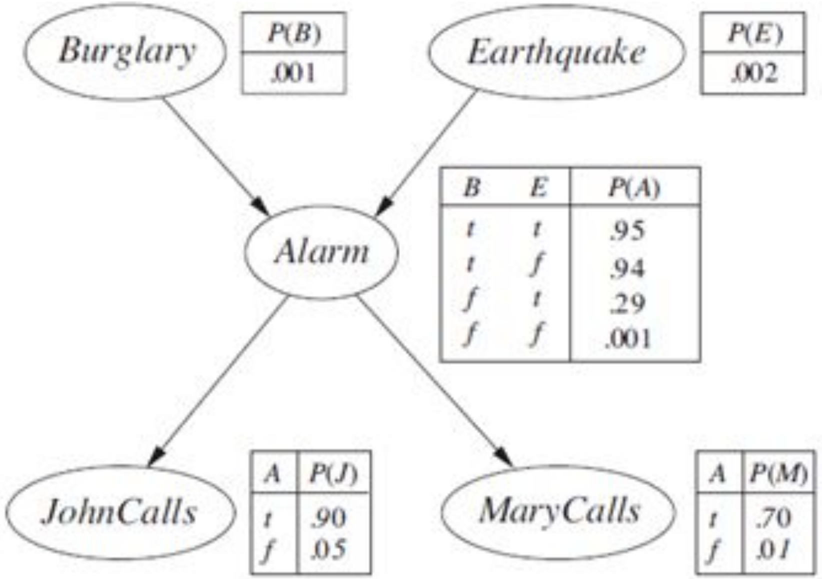
  - Required # of multiplications: 11, # of sums = 3, total = 14

- So order matters! However, finding optimal order is difficult!

Bayesian Network: Exact Inference

- Using the CPT data, we calculate:

$$P(b|j,m) = \alpha \times 0.00059224$$
$$P(\neg b|j,m) = \alpha \times 0.0014919$$



Hence,

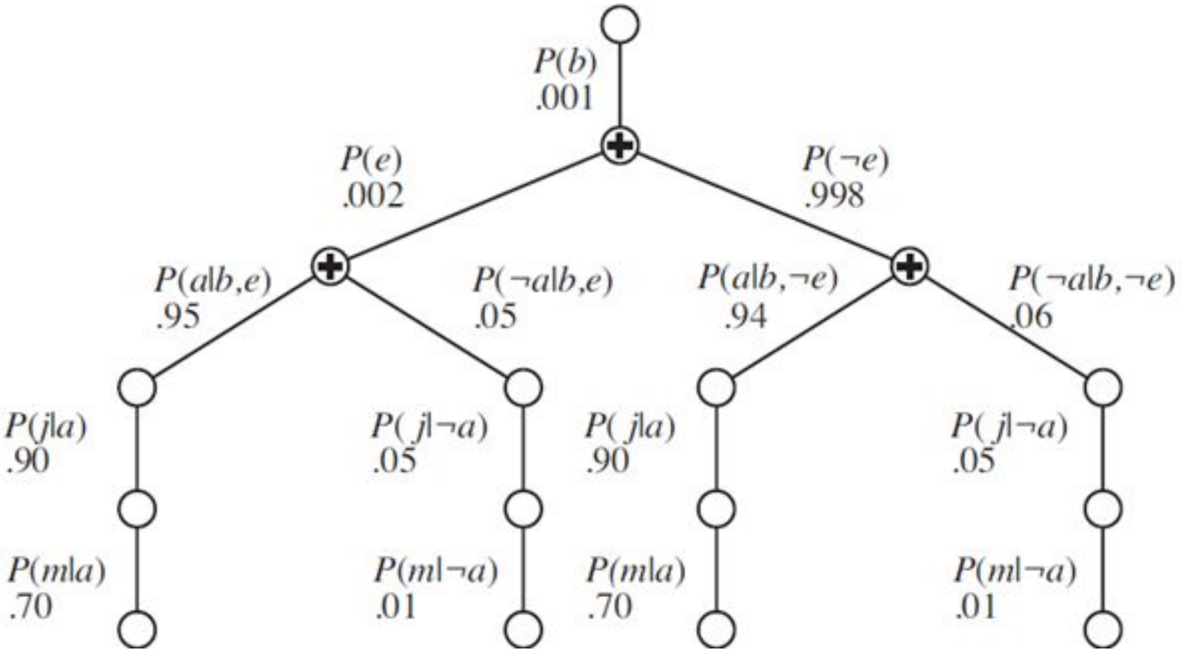
$$P(B|j,m) = \alpha \times (0.00059224, 0.0014919)$$
$$= (0.284, 0.716) \text{ [After normalizing as probabilities sum to 1]}$$

Bayesian Network: Exact Inference

- An evaluation tree is shown for the expression:

$$\alpha P(B) \sum_e P(e) \sum_a P(a|B,e) P(j|a) P(m|a)$$

  - Blank circled nodes represent multiplication [*Notice the repetition of sub-paths*]

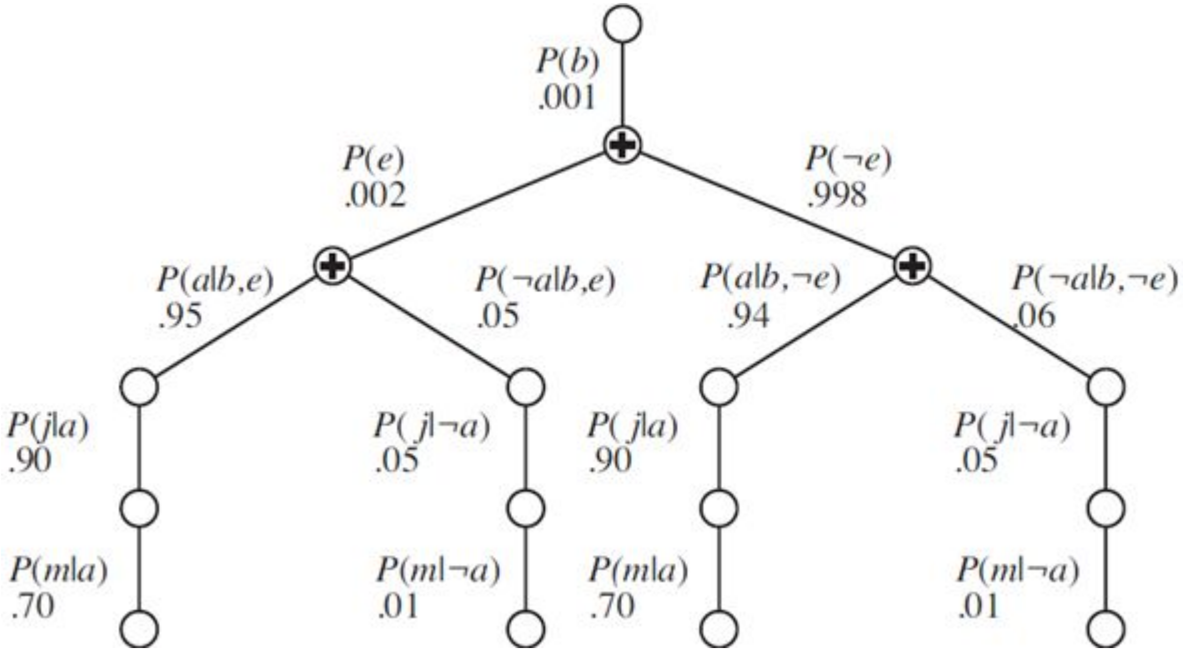


Bayesian Network: Exact Inference

- **Problem:** Repeated subexpression evaluation
- The following products are computed twice in two paths of the tree!

$$P(j \mid a)P(m \mid a)$$

$$P(j \mid \neg a)P(m \mid \neg a)$$



Bayesian Network: Variable Elimination Algorithm

- Steps of variable elimination algorithm:
  - Step 1: Compute factors for each node of Bayesian Network and instantiate/restrict evidence variables.
  - Step 2: For some order of variables $(V_1, V_2, .., V_N)$:
    - (2a) If variable V is a hidden variable, then multiply all the factors of V
    - (2b) Sum-out V
  - Step 3: Multiply remaining factors.
  - Step 4: Normalize the final factor to eliminate α.

Bayesian Network: Variable Elimination Algorithm

- Consider the following expression: $P(B|j,m)$.
  - Query variable: B
  - Evidence variable: J, M
  - Hidden variable: A, E

- We will use variable elimination algorithm to calculate

Bayesian Network: Variable Elimination Algorithm

- Query: $P(B|j, m)$.
- Joint probability expression of BN:

$$P(B, E, A, J, M) = P(B)P(E)P(A|B, E)P(J|A)P(M|A)$$

- **Step 1**: Compute all the factors of BN.
- The BN has five factors [*note the joint probability expression*]:
  - $f_1(B) = P(B)$
  - $f_2(E) = P(E)$
  - $f_3(A, B, E) = P(A|B, E)$
  - $f_4(J, A) = P(J|A)$
  - $f_5(M, A) = P(M|A)$

Variable Elimination Algorithm: Compute Factors

- Query: $P(B|j,m)$.
- **Step 1a**: Compute all the factors of BN. [*don't restrict evidences now*]

$f_1(B) = P(B)$

T	0.001
F	0.999

$f_2(E) = P(E)$

T	0.002
F	0.998

$f_3(A,B,E) = P(A|B,E)$

T	T	T	0.95
T	T	F	0.94
T	F	T	0.29
T	F	F	0.001
F	T	T	0.05
F	T	F	0.06
F	F	T	0.71
F	F	F	0.999

$f_4(J,A) = P(J|A)$

T	T	0.90
T	F	0.05
F	T	0.10
F	F	0.95

$f_5(M,A) = P(M|A)$

T	T	0.70
T	F	0.01
F	T	0.30
F	F	0.99

Variable Elimination Algorithm: Restrict Factors

- Query: $P(B|j, m)$.
- **Step 1b**: Instantiate/restrict evidence variables in all factors [j *and* m].
  - Restrict J *to* true *in* $f_4(J, A)$. [*Keep only rows where* $J = true$]
  - Note that J is removed from factor expression.

$f_4(J, A) = P(J|A)$

T	T	0.90
T	F	0.05
F	T	0.10
F	F	0.95

$f_4(A)$

T	0.90
F	0.05

Variable Elimination Algorithm: Restrict Factors

- Query: $P(B|j, m)$.
- **Step 1b**: Instantiate/restrict evidence variables in all factors. [j *and* m].
    - Restrict M *to* true *in* $f_5(M, A)$. [*Keep only rows where* $M = true$]

$f_5(M, A) = P(M|A)$

T	T	0.70
T	F	0.01
F	T	0.30
F	F	0.99

⇨

$f_5(A)$

T	0.70
F	0.01

Variable Elimination Algorithm: Multiply Factors

- Query: $P(B|j,m)$.
- **Step 2**: Select every variable V (in any order) which is hidden [Here, A and E]
  - Multiply all factors of V and then sum-out the variable.
  - Suppose we select in this order: E, A.
    - First we select E
    - Multiple all factors having E and get a new factor: $f_6(A,B,E) = f_2(E) \times f_3(A,B,E)$
    - Sum-out E and get a new factor: $f_7(A,B) = \sum_E f_6(A,B,E)$
    - Note that after summing-out, E is removed from factor-argument.

Bayesian Network: Multiply Factors

- **Step 2a**: Multiple all factors having E and get a new factor:

$$f_6(A,B,E) = f_2(E) \times f_3(A,B,E)$$

- Factor multiplication is a point-wise multiplication of elements in two factors:
  - Multiply rows which have matching values of the variables.
  - Number of rows in the new factor 2^N where N is number of total variables in the new factor:
    - Multiplying $f_1(A,B)$ and $f_2(B,C)$ will give us $f_3(A,B,C)$ which will have 8 rows.

Bayesian Network: Multiply Factors

- For variable E:
  - **Step 2a**: Multiple all factors having E and get a new factor:
  - **Multiply**: $f_6(A, B, E) = f_2(E) \times f_3(A, B, E)$

$f_2(E) = P(E)$

T	0.002
F	0.998

$\times$

$f_3(A, B, E) = P(A|B, E)$

T	T	T	0.95
T	T	F	0.94
T	F	T	0.29
T	F	F	0.001
F	T	T	0.05
F	T	F	0.06
F	F	T	0.71
F	F	F	0.999

$\Rightarrow$

$f_6(A, B, E)$

T	T	T	0.0019
T	T	F	0.93812
T	F	T	0.00058
T	F	F	0.000998
F	T	T	0.0001
F	T	F	0.05988
F	F	T	0.00142
F	F	F	0.997002

Bayesian Network: Multiply Factors

- For variable E:
  - **Step 2b**: Sum-out resulting factor for E and get a new factor:
  - **Sum-out *E***: $f_7(A,B) = \sum_E f_6(A,B,E)$ [*sum corresponding rows that have T and F for E*]

$f_6(A,B,E)$

T	T	T	0.0019
T	T	F	0.93812
T	F	T	0.00058
T	F	F	0.000998
F	T	T	0.0001
F	T	F	0.05988
F	F	T	0.00142
F	F	F	0.997002

$\Rightarrow$

$f_7(A,B)$

T	T	0.94002
T	F	0.001578
F	T	0.05998
F	F	0.998422

Bayesian Network: Multiply Factors

- For variable A:
  - **Step 2a**: Multiple all factors having A and get a new factor:
  - **Multiply**: $f_8(A,B) = f_4(A) \times f_5(A) \times f_7(A,B)$

$f_4(A)$

T	0.90
F	0.05

$\times$

$f_5(A)$

T	0.70
F	0.01

$\Rightarrow$

$f_4(A) \times f_5(A)$

T	0.63
F	0.0005

$\times$

$f_7(A,B)$

T	T	0.94002
T	F	0.001578
F	T	0.05998
F	F	0.998422

$\Rightarrow$

$f_8(A,B)$

T	T	0.592213
T	F	0.00099414
F	T	0.00002999
F	F	0.000499211

Bayesian Network: Multiply Factors

- For variable A:
  - **Step 2b**: Sum-out resulting factor for E and get a new factor:
  - **Sum-out *A***: $f_9(B) = \sum_A f_8(A,B)$ [*sum corresponding rows that have T and F for E*]

$f_8(A,B)$

T	T	0.592213
T	F	0.00099414
F	T	0.00002999
F	F	0.000499211

$\Rightarrow$

$f_9(B)$

T	0.59224299
F	0.001493351

Bayesian Network: Multiply Remaining Factors

- Query: $P(B|j,m)$.
- **Step 3:** Multiply remaining factors: $f_{10}(B) = f_1(B) \times f_9(B)$

$f_1(B) = P(B)$

T	0.001
F	0.999

$\times$

$f_9(B)$

T	0.59224299
F	0.001493351

$\Rightarrow$

$f_{10}(B)$

T	0.000592
F	0.001492

Bayesian Network: Normalize Result

- Query: $P(B|j, m)$.
- **Step 4**: Normalize the final factor to eliminate α.

$f_{10}(B)$

T	0.000592
F	0.001492

$\Rightarrow$

$f_{11}(B)$

T	0.284172
F	0.715828

- **Answer**: $P(b|m,j) = 0.284$, $P(\neg b|m,j) = 0.716$

Bayesian Network: Variable Elimination Algorithm

- Algorithm pseudocode [RN book]
  - Note: Step 1 (initialization) is missing in the pseudocode [it is implied in the MAKE-FACTOR function which is not a good approach!]

function ELIMINATION-ASK(X, $\mathbf{e}$, bn) **returns** a distribution over X
  inputs: X, the query variable
      $\mathbf{e}$, observed values for variables $\mathbf{E}$
      bn, a Bayesian network specifying joint distribution $\mathbf{P}(X_1, \ldots, X_n)$

  $factors \leftarrow [\,]$
  for each var **in** ORDER(bn.VARS) **do**
    $factors \leftarrow$ [MAKE-FACTOR(var, $\mathbf{e}$)|$factors$]
    if var is a hidden variable **then** $factors \leftarrow$ SUM-OUT(var, $factors$)
  return NORMALIZE(POINTWISE-PRODUCT($factors$))

Bayesian Network: Irrelevant Variable

- Consider the query: $P(JohnCalls|Burglary = true)$
- Incorporating all hidden variables of the Bayesian Network, we get:

$$P(J|b) = \alpha P(J,b) = \alpha \sum_{e} \sum_{a} \sum_{m} P(j,m,a,b,e) = \sum_{e} \sum_{a} \sum_{m} P(J|a)P(m|a)P(a|b,e)P(b)P(e)$$

Bayesian Network: Irrelevant Variable

- A simple rearrangement of the expression would yield us:

$$\mathbf{P}(J \mid b) = \alpha\, P(b) \sum_e P(e) \sum_a P(a \mid b, e) \mathbf{P}(J \mid a) \sum_m P(m \mid a)\,.$$

- Note that the inner term $\sum_m P(m|a)$ will evaluate to 1. So it is eliminated from the expression entirely [Irrelevant variable]

- *Variable elimination algorithm will automatically sum-out* m *from the expression!*
  - Why? [At some point, m will be considered as a hidden variable and the summation over m will eliminate it]

Bayesian Network: Irrelevant Variable

- In general, we can remove any leaf node that is not a query variable or an evidence variable.

- After removal of initial leaves, there may be some more leaf nodes, and these too may be irrelevant and removed [continue this process].

- *Every variable that is not an ancestor of a query variable or evidence variable is irrelevant to the query.*